Technical Requirement Document

Here's the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

Technical Requirement ID: TR-PS-001

Related Functional Requirement(s): FR-PS-001

Objective: Implement a PySpark job to read records from Oracle staging source, apply source extraction SQL, aggregate to derive source record count, and write the result to a flat file target.

- Target Cluster Configuration
- Cluster Name: Oracle Staging Data Processing Cluster
- Databricks Runtime Version: 7.3 LTS or higher
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (dependent on Autoscaling)
- Autoscaling: Enabled, Min Workers: 2, Max Workers: 5
- Auto Termination: Enabled, 30 minutes
- Libraries Installed:
- Oracle JDBC driver
- PySpark

- Source Data Details
- Dataset Name: STG_E2E_AC_TXDBH_DATA
- Location: Oracle staging database
- Format: Oracle table
- Description: Source data for reconciliation

- Target Data Details
- Output Name: FF_Source_Count
- Location: Designated flat file directory
- Format: Flat file (text)
- Description: Source record count output for reconciliation

- Job Flow / Pipeline Stages
- Read records from Oracle staging source using SQL defined by $$M_SRC_SQL

• Apply source extraction SQL to filter and extract required dataset

• Aggregate extracted dataset to derive SOURCE_REC_COUNT

• Write SOURCE_REC_COUNT to flat file target FF_Source_Count

• Data Transformations / Business Logic

• Step: Read Records from Oracle Staging Source

• Description: Read records from STG_E2E_AC_TXDBH_DATA using SQL defined by $$M_SRC_SQL

• Transformation Logic: Use Oracle JDBC driver to read data from Oracle staging source

• Step: Apply Source Extraction SQL

• Description: Apply SQL query defined in $$M_SRC_SQL to filter and extract required dataset

• Transformation Logic: Use PySpark to execute SQL query and extract dataset

• Step: Aggregate to Derive Source Record Count

• Description: Aggregate extracted dataset to derive SOURCE_REC_COUNT

• Transformation Logic: Use PySpark to count rows or count non-null request statuses

• Step: Write Result to Flat File Target

• Description: Write derived SOURCE_REC_COUNT to flat file target FF_Source_Count

• Transformation Logic: Convert SOURCE_REC_COUNT to string and write to flat file

• Error Handling and Logging

• Fail fast with clear error messaging if $$M_SRC_SQL fails to resolve or returns invalid SQL

• Log job execution details, including resolved SQL text/value for $$M_SRC_SQL and other relevant run identifiers

• Implement logging to track job execution and errors